

Linear Algebra 2

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Abstract

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How to relate different vector spaces.

6.1 Linear maps

Definition 6.1

Linear map

Let V and W be vector spaces over a field $\mathbb{F}$. A map $T : V \rightarrow W$ is linear if for all $u, v \in V$ and $\alpha \in \mathbb{F}$,

$$T(u + v) = T(u) + T(v), \quad T(\alpha v) = \alpha T(v).$$

Additivity and homogeneity

- Additivity: $T(u + v) = T(u) + T(v)$ for all $u, v \in V$.
- Homogeneity: $T(\alpha v) = \alpha T(v)$ for all $\alpha \in \mathbb{F}, v \in V$.

Notation

- $\mathcal{L}(V, W)$ = the space of linear maps $T : V \rightarrow W$
- $\mathcal{L}(V)$ = the space of linear maps $T : V \rightarrow V$

6.2 Examples

6.2.1 The zero linear map

$$\begin{aligned} 0 : V &\rightarrow W, \\ v &\mapsto 0. \end{aligned}$$

Remark

$$T(0_V) = 0_W.$$

$$\begin{aligned} 0_V &= 0_{\mathbb{F}}v, \\ T(0_V) &= T(0_{\mathbb{F}}v) = 0_{\mathbb{F}}T(v) = 0_W. \end{aligned}$$

6.2.2 The identity linear map

For any $v \in V$,

$$\begin{aligned} I : V &\rightarrow V, \\ v &\mapsto v. \end{aligned}$$

6.2.3 Differentiation operator

First recall the definition of a polynomial space over $\mathbb{R}$.

$$\mathcal{P}(\mathbb{R}) = \{\text{polynomials } a_0 + a_1x + a_2x^2 + \cdots + a_nx^n \mid a_i \in \mathbb{R}, n \in \mathbb{N}\}$$

Then define the differentiation operator.

$$\begin{aligned} D : \mathcal{P}(\mathbb{R}) &\rightarrow \mathcal{P}(\mathbb{R}) \\ \phi &\mapsto \phi' \end{aligned}$$

For $\phi \in \mathcal{P}(\mathbb{R})$,

$$\begin{aligned} \phi &= a_0 + a_1x + a_2x^2 + \cdots + a_nx^n, \\ \phi' &= a_1 + 2a_2x + \cdots + na_nx^{n-1}. \end{aligned}$$

For all $\alpha \in \mathbb{R}$ and all polynomials $p, q \in \mathcal{P}(\mathbb{R})$,

$$\begin{aligned} (p + q)' &= p' + q', \\ (\alpha p)' &= \alpha p'. \end{aligned}$$

The differentiation operator is linear.

6.2.4 Integration operator

Define the integration operator.

$$\begin{aligned}\int: \mathcal{P}(\mathbb{R}) &\rightarrow \mathcal{P}(\mathbb{R}) \\ \phi &\mapsto \int \phi \, dx\end{aligned}$$

For all $\alpha \in \mathbb{R}$ and all polynomials $p, q \in \mathcal{P}(\mathbb{R})$,

$$\begin{aligned}\int(p+q) &= \int p + \int q, \\ \int(\alpha p) &= \alpha \int p.\end{aligned}$$

6.2.5 Multiplication by a fixed polynomial

Define the multiplication by a fixed polynomial operator.

$$\begin{aligned}T: \mathcal{P}(\mathbb{R}) &\rightarrow \mathcal{P}(\mathbb{R}) \\ \phi &\mapsto q\phi\end{aligned}$$

So

$$(Tp)(x) = q(x)p(x).$$

Check that T is linear.

6.2.6 Backward shift operator

Define the backward shift operator.

$$\begin{aligned}T: \mathbb{F}^\infty &\rightarrow \mathbb{F}^\infty \\ (x_1, x_2, \dots) &\mapsto (x_2, x_3, \dots)\end{aligned}$$

Check that T is linear.

6.2.7 Forward shift operator

Define the forward shift operator.

$$\begin{aligned}T: \mathbb{F}^\infty &\rightarrow \mathbb{F}^\infty \\ (x_1, x_2, \dots) &\mapsto (0, x_1, x_2, \dots)\end{aligned}$$

Check that T is linear.

6.2.8 Linear map and matrices

$$\begin{aligned}T: \mathbb{F}^n &\rightarrow \mathbb{F}^m, \\ \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} &\mapsto \begin{bmatrix} A_{11}x_1 + A_{12}x_2 + \cdots + A_{1n}x_n \\ A_{21}x_1 + A_{22}x_2 + \cdots + A_{2n}x_n \\ \vdots \\ A_{m1}x_1 + A_{m2}x_2 + \cdots + A_{mn}x_n \end{bmatrix}.\end{aligned}$$

T should be a matrix.

$$\begin{bmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} & A_{m2} & \dots & A_{mn} \end{bmatrix}_{(m \times n)} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$m \times n$ matrix: rows $\times$ columns.

$$T \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, T \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \dots, T \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

Columns!

So, because the lemma below shows that a linear map is completely determined by its action on the basis vectors of the domain. Any linear map can be uniquely represented by a matrix with respect to the chosen bases.

6.3 A linear map is determined by its values on a basis

Lemma 6.1

A linear map $T: V \rightarrow W$ is uniquely determined by what it does on a basis of V .

Example

For a basis $\{v_1, \dots, v_n\}$ of V , let

$$T(v_i) = w_i \in W \quad \text{for } i = 1, 2, \dots, n.$$

$$T(v_1) = w_1,$$

$$T(v_2) = w_2,$$

$$\vdots$$

$$T(v_n) = w_n.$$

Then T is uniquely determined by the images of the basis vectors.

$$\begin{bmatrix} w_1 & w_2 & \dots & w_n \end{bmatrix}$$

determines T uniquely. For $v \in V$,

$$\begin{aligned} Tv &= T(a_1v_1 + a_2v_2 + \dots + a_nv_n) \\ &= a_1T(v_1) + a_2T(v_2) + \dots + a_nT(v_n) \\ &= a_1w_1 + a_2w_2 + \dots + a_nw_n. \end{aligned}$$

Check that T is a linear map.

6.3.1 $\mathcal{L}(V, W)$

Let $S, T \in \mathcal{L}(V, W)$.

$$\begin{aligned} (S + T)(v) &= S(v) + T(v) && \text{for all } v \in V, \\ (\lambda S)(v) &= \lambda S(v) && \text{for all } v \in V \text{ and } \lambda \in \mathbb{F}. \end{aligned}$$

Claim (Need to check)

$S + T, \lambda S$ are linear maps.

That makes $\mathcal{L}(V, W)$ a vector space over $\mathbb{F}$.

Similarly, $\mathcal{L}(V)$ is a vector space over $\mathbb{F}$ for any vector space V .

6.3.2 Compositions of linear maps

Case 1 Let U, V, W be vector spaces over $\mathbb{F}$.

$$\begin{array}{ccccc} U & \xrightarrow{S} & V & \xrightarrow{T} & W & \xrightarrow{R} & X \\ & & & \nearrow^{R \circ T} & & & \\ & \searrow_{T \circ S} & & & & & \end{array}$$

For all $u \in U$,

$$\begin{aligned} (T \circ S)(u) &= T(S(u)), \\ (R \circ (T \circ S))(u) &= R(T(S(u))), \\ ((R \circ T) \circ S)(u) &= (R \circ T)(S(u)) = R(T(S(u))). \end{aligned}$$

Therefore,

$$R \circ (T \circ S) = (R \circ T) \circ S.$$

Case 2 For $T: V \rightarrow W$.

$$\begin{array}{ccccc} V & \xrightarrow{I_V} & V & \xrightarrow{T} & W & \xrightarrow{I_W} & W \\ & & & \nearrow^T & & & \\ & \searrow_T & & & & & \end{array}$$

So

$$\begin{aligned} T \circ I_V &= T, \\ I_W \circ T &= T. \end{aligned}$$

For $S, T: V \rightarrow V$, the compositions $S \circ T$ and $T \circ S$ need not agree.

Example

For $V = \mathcal{P}(\mathbb{R})$,

$$T: \mathcal{P}(\mathbb{R}) \rightarrow \mathcal{P}(\mathbb{R})$$

$$p(x) \mapsto p'(x)$$

$$S: \mathcal{P}(\mathbb{R}) \rightarrow \mathcal{P}(\mathbb{R})$$

$$p(x) \mapsto xp(x)$$

$$T(p(x)) = p'(x), S(p(x)) = xp(x).$$

$$(S \circ T)(p(x)) = S(p'(x)) = xp'(x),$$

$$(T \circ S)(p(x)) = T(xp(x)) = p(x) + xp'(x).$$

6.4 Kernel and Image of a linear map

Definition 6.2

Kernel and Image Spaces

Let $T: V \rightarrow W$ be a linear map.

$$\ker T = \{v \in V \mid T(v) = 0_W\} \quad \text{kernel of } T \text{ (null space of } T),$$

$$\text{Im } T = \{T(v) \mid v \in V\} \quad \text{image of } T \text{ (range of } T).$$

Example

Zero map For the zero map,

$$0: V \rightarrow W$$

$$v \mapsto 0$$

We can immediately see that the range of the zero map is just the zero vector in W , and the null space is the entire space V .

$$\text{Im}(0) = \{0\},$$

$$\ker(0) = V.$$

Example

Differentiation map For the differentiation map,

$$D: \mathcal{P}(\mathbb{R}) \rightarrow \mathcal{P}(\mathbb{R})$$

$$\gamma \mapsto \gamma'$$

We can immediately see that the range of the differentiation map is the set of all derivatives of functions in $\mathcal{P}(\mathbb{R})$, and the null space is the set of all constant polynomials.

$$\text{Im}(D) = \{\gamma' \mid \gamma \in \mathcal{P}(\mathbb{R})\},$$

$$\ker(D) = \{\text{constant polynomials}\}.$$

7 Lecture 7

7.1 Properties of Linear Maps

Let V and W be vector spaces over the same field $\mathbb{F}$. Let $T: V \rightarrow W$ be a linear map. For all $v_1, v_2 \in V$ and $\alpha \in \mathbb{F}$,

$$\begin{cases} T(v_1 + v_2) = T(v_1) + T(v_2), \\ T(\alpha v_1) = \alpha T(v_1), \end{cases} \iff T(u + \alpha v) = T(u) + T(\alpha v) = T(u) + \alpha T(v).$$

7.2 Injective and Surjective Maps

7.2.1 Kernel and Image Definition

Definition 7.1

The kernel of T

Let $T: V \rightarrow W$ be a linear map.

$$\ker T = \{v \in V \mid T(v) = 0_W\}.$$

Definition 7.2

The image of T

Let $T: V \rightarrow W$ be a linear map.

$$\operatorname{Im} T = \{T(v) \mid v \in V\}.$$

Theorem 7.1

The kernel and image of a linear map T are vector subspaces of V and W , respectively.

$$\ker T \subseteq V,$$

$$\operatorname{Im} T \subseteq W.$$

Proof. Since $T(0_V) = 0_W$, the kernel contains 0_V .

If $v_1, v_2 \in \ker T$ and $\lambda \in \mathbb{F}$, then

$$T(v_1 + \lambda v_2) = T(v_1) + \lambda T(v_2) = 0_W,$$

so $v_1 + \lambda v_2 \in \ker T$.

Thus $\ker T$ is a subspace of V . □

Proof. If $w_1, w_2 \in \operatorname{Im} T$, write

$$w_1 = T(v_1),$$

$$w_2 = T(v_2)$$

for some $v_1, v_2 \in V$. For $\lambda \in \mathbb{F}$,

$$w_1 + \lambda w_2 = T(v_1) + \lambda T(v_2) = T(v_1 + \lambda v_2) \in \operatorname{Im} T.$$

Thus $\operatorname{Im} T$ is a subspace of W . □

7.2.2 Injective

Definition 7.3

The linear map $T: V \rightarrow W$ is called *injective* (or one-to-one) if $\ker T = \{0_V\}$.

For $T: V \rightarrow W$, injectivity is equivalent to

$$T(v_1) = T(v_2) \implies v_1 = v_2.$$

Indeed, if $T(v_1) = T(v_2)$, then

$$T(v_1 - v_2) = 0_W,$$

so

$$v_1 - v_2 \in \ker T = \{0_V\},$$

and hence

$$v_1 = v_2.$$

Conversely, if T is injective and $v \in \ker T$, then

$$T(v) = T(0_V),$$

so $v = 0_V$.

7.2.3 Surjective

Definition 7.4

The linear map $T: V \rightarrow W$ is called *surjective* (or onto) if $\operatorname{Im} T = W$.

If T is surjective, then by definition $\operatorname{Im} T = W$.

Equivalently, for every $w \in W$ there exists $v \in V$ such that

$$w = T(v).$$

Example

The derivative map

$$\begin{aligned} D: \mathcal{P}(\mathbb{R}) &\rightarrow \mathcal{P}(\mathbb{R}) \\ p &\mapsto p' \end{aligned}$$

has

$$\begin{aligned} \ker D &= \{p \in \mathcal{P}(\mathbb{R}) \mid D(p) = 0\}, \\ \operatorname{Im} D &= \{D(p) \mid p \in \mathcal{P}(\mathbb{R})\}. \end{aligned}$$

D is surjective but not injective.

Example

The backward shift b and forward shift f on $\mathbb{F}^\infty$ are defined by

$$\mathbb{F}^\infty \begin{matrix} \xrightarrow{b} \\ \xrightarrow{f} \end{matrix} \mathbb{F}^\infty$$

$$\begin{aligned} b(x_1, x_2, x_3, \dots) &= (x_2, x_3, x_4, \dots), \\ f(x_1, x_2, x_3, \dots) &= (0, x_1, x_2, \dots). \end{aligned}$$

The kernels and images of the backward and forward shifts are given by

$$\begin{aligned} \ker b &= \{(x_1, x_2, x_3, \dots) \in \mathbb{F}^\infty \mid x_2 = x_3 = \dots = 0\}, \\ \operatorname{Im} b &= \mathbb{F}^\infty, \\ \ker f &= \{0\}, \\ \operatorname{Im} f &= \{(x_1, x_2, x_3, \dots) \in \mathbb{F}^\infty \mid x_1 = 0\}. \end{aligned}$$

The backward shift is surjective but not injective, while the forward shift is injective but not surjective.

Example

$$\begin{aligned} T: \mathcal{P}(\mathbb{R}) &\rightarrow \mathcal{P}(\mathbb{R}), \\ p(x) &\mapsto xp(x). \end{aligned}$$

$$\ker T = \{p(x) \in \mathcal{P}(\mathbb{R}) \mid xp(x) = 0\} = \{0\}.$$

$$p(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n, \quad a_i \in \mathbb{R}.$$

$$xp(x) = a_0x + a_1x^2 + a_2x^3 + \dots + a_nx^{n+1}.$$

$$\operatorname{Im} T = \{xp(x) \mid p(x) \in \mathcal{P}(\mathbb{R})\} \neq \mathcal{P}(\mathbb{R}).$$

T is injective but not surjective.

7.3 The Fundamental Theorem of Linear maps

Theorem 7.2

The fundamental theorem of linear maps (also known as the rank-nullity or rank theorem)

If $T: V \rightarrow W$ is a linear map, and the dimension of V is finite, then

$$\dim V = \underbrace{\dim \ker T}_{\text{nullity}} + \underbrace{\dim \operatorname{Im} T}_{\text{rank}}.$$

Proof. Since $\ker T \subseteq V$, let $\{u_1, \dots, u_n\}$ be a basis for $\ker T$. Extend it to a basis for V .

$$\{u_1, \dots, u_n, v_1, \dots, v_r\}$$

Thus

$$\dim V = n + r, \quad \dim \ker T = n.$$

We want to show that $\dim \operatorname{Im} T = r$, so we claim that $\{T(v_1), \dots, T(v_r)\}$ is a basis for $\operatorname{Im} T$, need to be proven.

Claim: $\{T(v_1), \dots, T(v_r)\}$ is a basis for $\text{Im } T$.

1. $\{T(v_1), \dots, T(v_r)\}$ spans $\text{Im } T$.

Since $\{u_1, \dots, u_n, v_1, \dots, v_r\}$ is a basis for V , any $v \in V$ can be written as

$$v = a_1u_1 + \dots + a_nu_n + b_1v_1 + \dots + b_rv_r.$$

Applying T , we get

$$T(v) = b_1T(v_1) + \dots + b_rT(v_r),$$

since $T(u_i) = 0$ for all i . Therefore, $\{T(v_1), \dots, T(v_r)\}$ spans $\text{Im } T$.

2. $\{T(v_1), \dots, T(v_r)\}$ is linearly independent.

Suppose

$$c_1T(v_1) + \dots + c_rT(v_r) = 0.$$

Then

$$T(c_1v_1 + \dots + c_rv_r) = 0.$$

Therefore, $c_1v_1 + \dots + c_rv_r \in \ker T$. But $\{u_1, \dots, u_n, v_1, \dots, v_r\}$ is a basis for V , so $c_1v_1 + \dots + c_rv_r$ can be written uniquely as a linear combination of the basis vectors:

$$c_1v_1 + \dots + c_rv_r = a_1u_1 + \dots + a_nu_n.$$

Hence

$$c_1v_1 + \dots + c_rv_r - (a_1u_1 + \dots + a_nu_n) = 0.$$

Since the basis vectors are linearly independent, we must have

$$c_1 = \dots = c_r = 0.$$

This proves that $\{T(v_1), \dots, T(v_r)\}$ is linearly independent, completing the proof of the claim. Therefore, $\{T(v_1), \dots, T(v_r)\}$ is a basis for $\text{Im } T$.

$$\dim \text{Im } T = r, \quad \dim \ker T = n, \quad \dim V = n + r.$$

$$a_1 = \dots = a_n = 0, \quad c_1 = \dots = c_r = 0.$$

This proves that $\{T(v_1), \dots, T(v_r)\}$ is linearly independent, completing the proof of the claim.

Therefore, $\{T(v_1), \dots, T(v_r)\}$ is a basis for $\text{Im } T$.

$$\dim \text{Im } T = r, \quad \dim \ker T = n, \quad \dim V = n + r.$$

□

7.4 Corollaries of the Rank Theorem

Let $T: V \rightarrow W$ with

1. $\dim V < \infty$
2. $\dim W < \infty$.

7.4.1 Injective Maps

If T is injective, then

$$\dim V = \dim \operatorname{Im} T \leq \dim W,$$

because $\ker T = \{0_V\}$.

So if $\dim V > \dim W$, no injective map $T: V \rightarrow W$ exists.

7.4.2 Surjective Maps

If T is surjective, then

$$\dim W = \dim \operatorname{Im} T \leq \dim V,$$

because $\operatorname{Im} T = W$.

So if $\dim V < \dim W$, no surjective map $T: V \rightarrow W$ exists.

7.4.3 Bijective Maps

If $T: V \rightarrow W$ is injective and surjective, then $\dim V = \dim W$.

Proof. Since T is injective,

$$\dim V = \dim \operatorname{Im} T \leq \dim W.$$

Since T is surjective,

$$\dim W = \dim \operatorname{Im} T \leq \dim V.$$

Therefore,

$$\dim V = \dim W.$$

□